Solid-waste management involves the collection, treatment and proper disposing of solid material that is discarded from the household and industrial activities.

Methods of solid wastes disposal

(i) **Segregation**: It is the separation of different type of waste materials like biodegradable and non biodegradable wastes.

(ii) **Sanitary landfill**: Solid wastes are dumped into low lying areas. The layers are compacted by trucks to allow settlement. The waste materials get stabilised in about 2-12 months. The organic matter undergoes decomposition.



Figure 22.6 Collection of degradable and non-degradable solid wastes

(iii) **Incineration**: It is the burning of non-biodegradable solid wastes (medical wastes) in properly constructed furnace at high temperature.

(iv) **Composting**: Biodegradable matter of solid wastes is digested by microbial action or earthworms and converted into humus.



Figure 22.7 Collection of various types of solid wastes in separate bins

Recycling of wastes

- Papers from old books, magazines and newspapers are recycled to produce papers in papermills.
- Agricultural wastes like coconut shells, jute cotton stalk, bagasse of sugarcane can be used to make paper and hard board. Paddy husk can be used as livestock fodder.
- Cowdung and other organic wastes can be used in gobar gas plant to provide biogas and manure for fields.

4R Approach

The 4R approach such as Reduce, Reuse, Recovery and Recycle may be followed for effective waste management.

Points to Remember

- Conservation is a process which is concerned with the use, preservation and proper management of natural resources from destructive activities of human being.
- Conservation of natural resources contributes to the social and economic development of the country.
- Forests of a country constitute a major asset for the people of the country.
- National park is a reserved area for the conservation of entire wildlife including plants and animals.
- Sanctuary is a place reserved exclusively for the use of animals.
- Solar cell is a device that absorbs sunlight and converts it into electric energy.
- Solar water heater does not require electricity, they heat up water directly from sunlight.
- Biogas is produced by the anaerobic decomposition of cow dung.
- The technique of collecting and storing rain water for future purpose is known as rainwater harvesting.
- Unwanted, non-working and outdated electronic products become e-waste.

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